INCH-POUND
MIL-PRF-3098/24E
9 December 2004
SUPERSEDING
MIL-PRF-3098/24D
27 August 1997

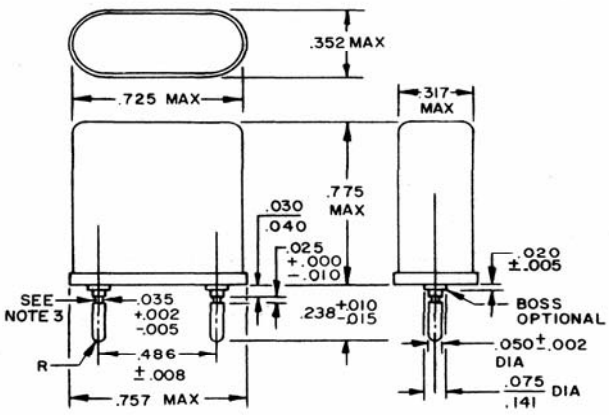
PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR45/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification and MIL-PRF-3098.

Pertinent characteristics: 455 kHz; fundamental; noncontrolled; series resonance.



Inches	mm	Inches	mm	Inches	mm	Inches	mm
.002	0.05	.020	0.51	.050	1.27	.352	8.94
.005	0.13	.025	0.64	.075	1.91	.486	12.34
.008	0.20	.030	0.76	.141	3.58	.725	18.42
.010	0.25	.035	0.89	.238	6.05	.757	19.23
.015	0.38	.040	1.02	.317	8.05	.775	38.76

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. The pin undercut may be omitted.
4. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR45/U.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 455 kHz.

Capacitance, shunt: 5.0 pF ±2.5 pF.

Mode of oscillation: Fundamental.

Rated drive level: 1.0 mW, maximum.

Resonance: Series.

Calibration values:

Equivalent resistance: 2,300 ohms.

Resistor voltage drop: 2.14 volts.

Operating temperature range (noncontrolled): -40°C to +70°C, inclusive.

Frequency tolerance: ±200 parts per million (ppm). There shall be a difference in frequency of 80 ±12 Hertz between operation at series resonance and operation at antiresonance into load capacitance of 32.0 pF ±0.5 pF. Both measurements for this purpose shall be made at room temperature.

Equivalent resistance: 3,300 ohms, maximum.

Shock (specified pulse):

Frequency change permitted: ± 30 ppm.

Equivalent resistance change permitted: ±15 percent.

Vibration: Method 201 of MIL-STD-202, amplitude .015 inch (0.38 mm) (.030 inch (0.76 mm) total excursion).

Frequency change permitted: ±30 ppm.

Equivalent resistance change permitted: ±15 percent.

Temperature run:

Frequency change permitted: ±10 ppm.

Equivalent resistance change permitted: ±15 percent.

Bond strength: 500 grams, minimum.

* Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202

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* The margins of this specification sheet are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - EC
Air Force - 11
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(Project 5955-0773)

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